ARTICLE | VIROLOGY | AUTOMATED ANALYSIS

Within-host diversity of wheat streak mosaic virus is not detectably altered by coinfection with Triticum mosaic virus

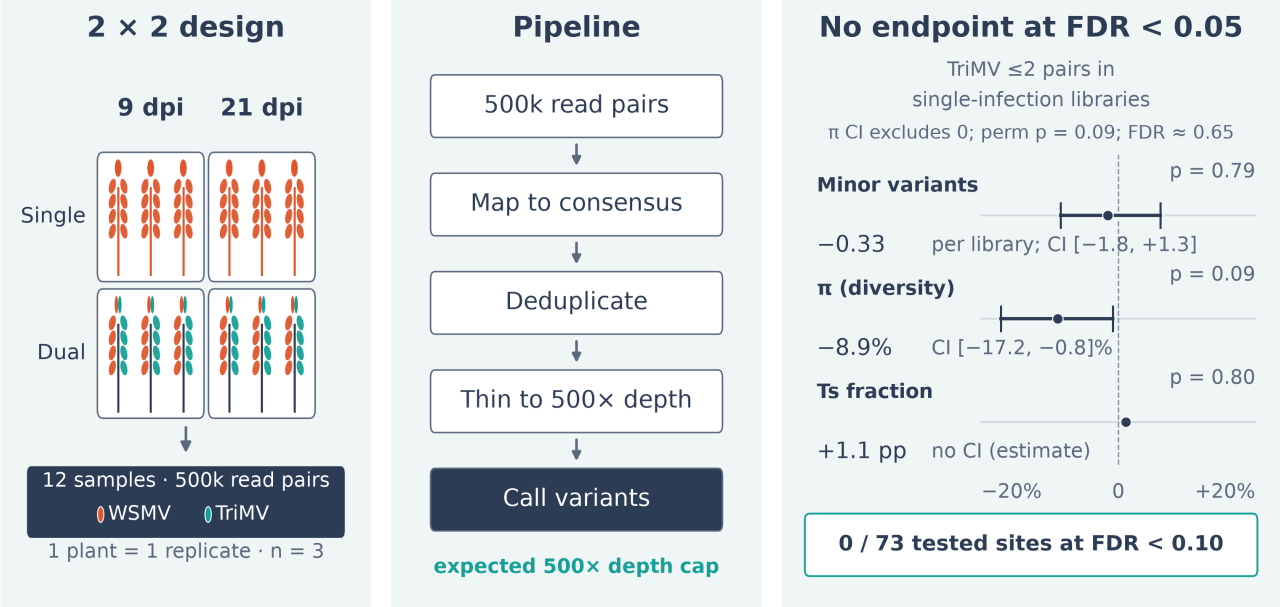
OpenBioWorkstation 课题组（自动化分析）

Automated analysis performed with OpenBioWorkstation. No individual authors are listed; correspondence is not applicable.

HIGHLIGHTS

- Twelve wheat libraries reanalyzed at a matched expected depth in a 2×2 design
- Low-frequency WSMV variants are common in every treatment group
- Coinfection effects on WSMV diversity are small and not significant
- No site passes FDR < 0.10; results hold across depth and MAF grids

eTOC blurb: Reanalyzing 12 wheat RNA-seq libraries at a matched expected depth, we find that coinfection with Triticum mosaic virus does not detectably change WSMV minor-variant counts, nucleotide diversity, or substitution spectra. The result is stable across rarefaction depth, minor-allele-frequency threshold, and read support; with three replicates per cell, confidence intervals remain wide and do not establish equivalence.



Graphical abstract. Expected-depth comparison of WSMV diversity in single versus dual infection. Effects are dual minus single on a shared relative scale.

SUMMARY

Wheat streak mosaic virus (WSMV) and Triticum mosaic virus (TriMV) frequently coinfect wheat, but whether coinfection reshapes WSMV within-host diversity is unknown. We reanalyzed 12 RNA-seq libraries from a 2×2 design at 500,000 read pairs per sample, mapping reads to a consensus reference and applying expected-depth binomial thinning to a 500× cap. WSMV minor variants were abundant in every group (group medians 19–22 per sample). The dual-minus-single

effect was −0.33 variants (95% CI −1.83 to 1.34; permutation p = 0.79), and no tested candidate site was significant after correction (0 of 73 sites at FDR < 0.10; 31 recurrent sites). These data provide no evidence that TriMV coinfection detectably alters WSMV within-host diversity in these samples.

Keywords: wheat streak mosaic virus; Triticum mosaic virus; mixed infection; within-host diversity; minor variants; RNA-seq; rarefaction

Introduction

Wheat streak mosaic virus (WSMV, genus *Tritimovirus*) and Triticum mosaic virus (TriMV, genus *Poacevirus*) are among the most damaging viruses of wheat in the Great Plains of North America. Both are transmitted by the wheat curl mite *Aceria tosichella*, and mixed infections are common in the field [1, 3, 10]. In controlled experiments, coinfection can increase viral accumulation in a cultivar-dependent manner, with WSMV titers reported to rise several-fold in some wheat cultivars [3, 5]. We did not find a direct comparison of whether such synergistic changes in accumulation are accompanied by changes in the genetic composition of the WSMV population within a plant in the literature searches recorded for this study (limited to the databases, terms, and date recorded in analysis/lit_search_log.tsv).

WSMV populations are typically described as genetically narrow. Clone-based studies of serial-passage lineages estimated mutation frequencies near 5×10^{-4} per nucleotide and inferred strong population bottlenecks during systemic movement [1, 2]. TriMV likewise shows limited population variation after repeated passage [4]. These observations have framed WSMV as a virus whose diversity is constrained by founder effects rather than by sustained diversifying selection. However, clone-based surveys sample a small number of genome segments and cannot resolve low-frequency variants across the whole genome. High-throughput sequencing (HTS) has since been used to describe WSMV and TriMV diversity at the field and evolutionary scale [6, 8, 9], but these studies compare isolates or geographic regions rather than single versus mixed infection within a common host background.

Coinfection could alter within-host diversity through several non-exclusive routes. Increased viral accumulation may increase the number of replication templates and thus the supply of new mutations; competition for shared host factors may instead narrow the population; and heterologous helper functions may relax bottlenecks during cell-to-cell or systemic movement. These routes predict different signs of effect, so the direction is an empirical question. Answering it requires (i) single- and dual-infection samples generated in the same host genotype and environment, (ii) whole-genome rather than amplicon data, and (iii) equal effective depth, because variant detection is strongly depth-dependent and viral read fractions differ among libraries.

The public resource PRJNA1165236 provides exactly this design: wheat cv. Tomahawk plants infected with WSMV Sidney 81 alone or together with TriMV Nebraska, sampled at 9 and 21 days post-inoculation (dpi) with three biological replicates per treatment [7]. The original study analyzed the host transcriptome and did not examine viral minor variants [7]. Here we reuse 12 of these libraries, compare WSMV populations at matched effective depth, and ask whether TriMV coinfection is associated with a detectable change in WSMV minor-variant burden, nucleotide diversity, or substitution spectrum.

Results

Sequencing depth and virome composition

We selected 12 libraries comprising the WSMV single-infection and WSMV+TriMV dual-infection arms at 9 and 21 dpi (three replicates each) and subsampled 500,000 read pairs per library. After adapter and quality trimming, reads were mapped to a consensus reference in which the single fixed TriMV difference (position 8233, T → A; depth 1,215, frequency 1.0000) was incorporated, so that the reference did not impose a known mismatch on all libraries. Duplicates were marked and excluded from pileup; paired-end overlap clipping was applied by htslib at the pileup stage, and a coordinate-based overlap coefficient was recorded only as a diagnostic.

Viral read fractions differed among groups but were sufficient for deep, genome-wide coverage (Table 1 and Figure 1). WSMV mean depth ranged from 1,147× to 2,798× across the 12 libraries; TriMV mean depth in dual infections ranged from 913× to

1,196×. The fraction of WSMV sites covered at ≥400× was 98.4–99.2% in all groups, and the median non-reference base background rate was 4.96–5.77×10^{-4} per base (Table 1); this background includes real low-frequency variants and library or mapping errors, not only sequencing error. TriMV was essentially absent from single-infection libraries (≤2 read pairs per library), as expected. Read fractions are compositional and are not interpreted as absolute viral loads.

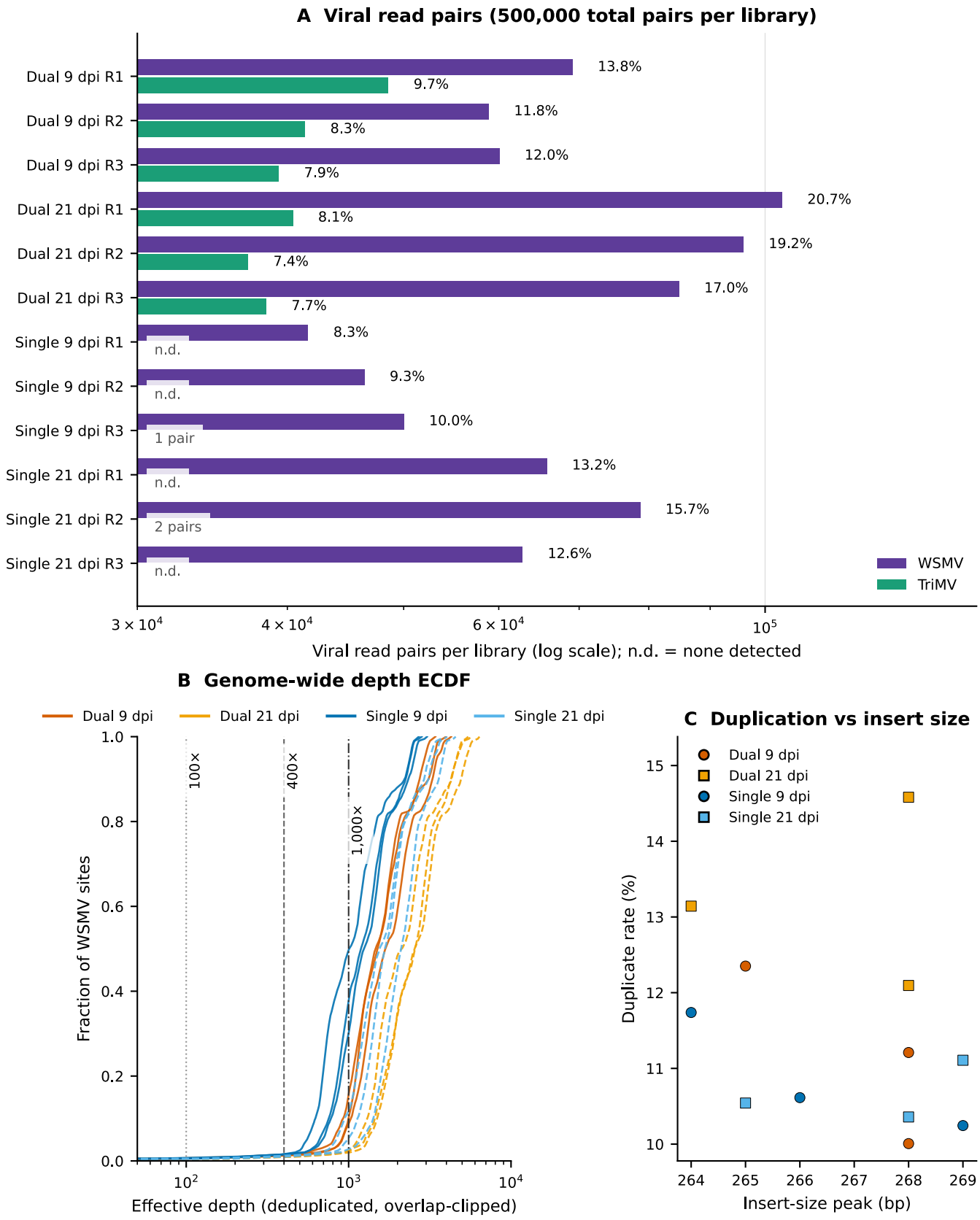


Figure 1. Viral read composition and sequencing depth. (A) WSMV and TriMV read pairs per library on a log scale; percentages are of 500,000 total pairs; n.d., none detected. (B) ECDF of effective WSMV depth (deduplicated, overlap-clipped) with 100×, 400×, and 1,000× reference lines; median breadth at these thresholds was 0.994/0.986/0.898 for dual 9 dpi, 0.997/0.991/0.976 for dual 21 dpi, 0.995/0.984/0.622 for single 9 dpi, and 0.996/0.989/0.946 for single 21 dpi. (C) Library duplicate rate versus insert-size peak; color denotes group, shape denotes 9 or 21 dpi. Read fractions are compositional and do not estimate absolute viral load.

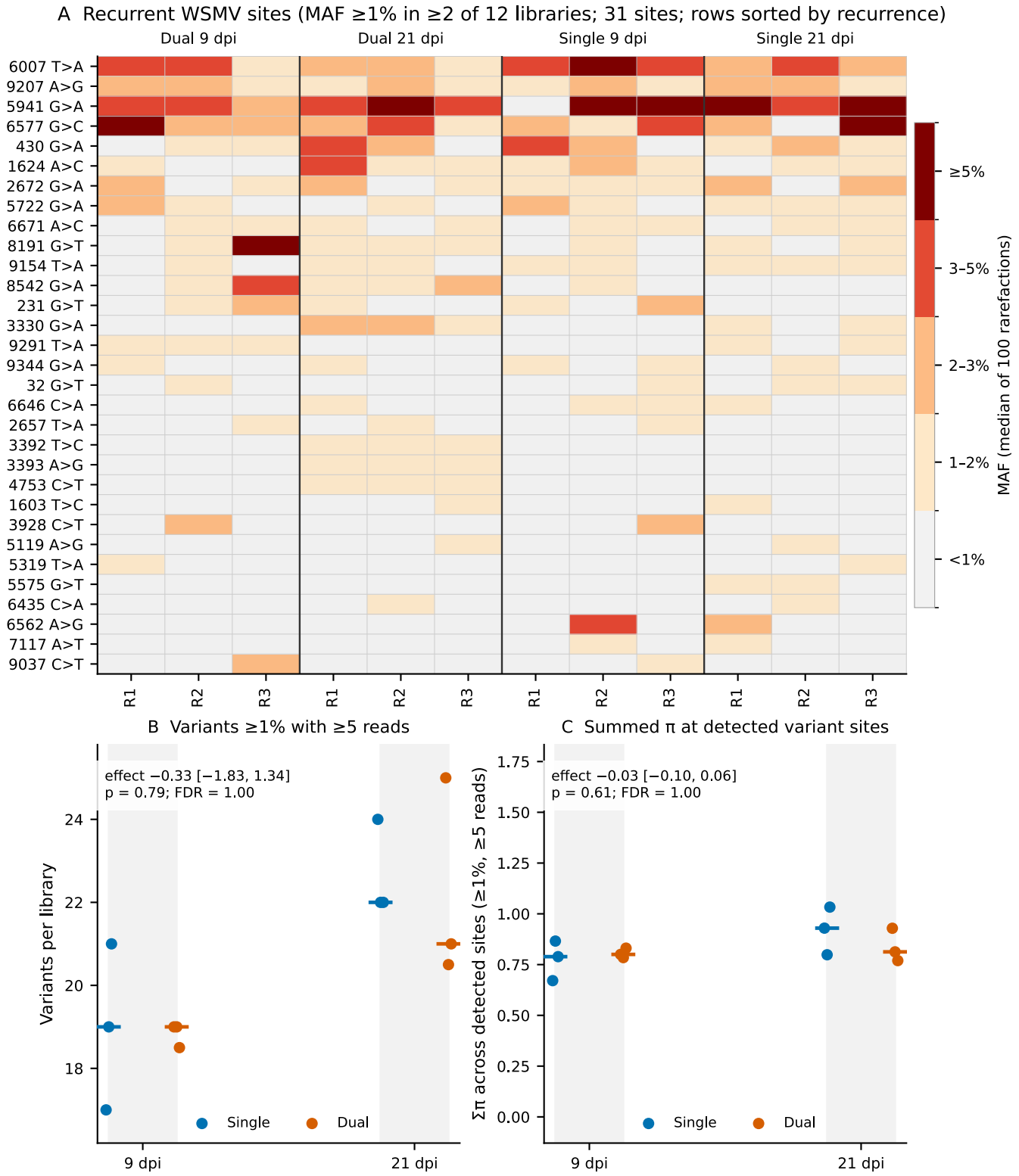


Figure 2. Recurrent WSMV minor variants and sample-level effects. (A) Heatmap of 31 WSMV sites with MAF ≥1% in at least 2 of 12 libraries; rows are sorted by recurrence; light gray denotes MAF <1%; cell values are medians over 100 thinning replicates at a 500× cap. These recurrent sites are a subset of the 73 candidate sites tested in the site-level analysis. Black vertical lines separate the four groups. (B) Number of variants at ≥1% MAF with ≥5 supporting reads per library; horizontal bars are group medians. (C) Summed π across detected variant sites; this metric can exceed 1 because it sums per-site values. Effect, 95% bootstrap CI, permutation p, and Benjamini–Hochberg FDR (seven-endpoint family) are shown in each panel. No connecting lines are drawn because replicate numbers are not paired across treatments.

Table 1. Group-level sequencing and diversity summary (median of three replicates).

Group	WSMV pairs (%)	WSMV depth (×)	⩾400× breadth	Variants ⩾1%	π (×10^{-3})	Σπ detected
Dual, 9 dpi	11.8–13.8	1,696	0.986	19.0	1.01	0.80
Dual, 21 dpi	17.0–20.7	2,623	0.991	21.0	1.10	0.81
Single, 9 dpi	8.3–10.0	1,308	0.985	19.0	1.17	0.79
Single, 21 dpi	12.6–15.7	1,807	0.988	22.0	1.16	0.93

Variants: ≥1% minor-allele frequency (MAF) with ≥5 supporting reads, median over 100 expected-depth thinning replicates at a 500× cap. π: mean per-site nucleotide diversity over covered sites. Σπ detected: sum of per-site π over detected variant sites. Pair fractions are of 500,000 total pairs and are not absolute loads.

Minor-variant burden at the expected-depth cap

At the 500× expected-depth cap, every library carried low-frequency WSMV variants. The median number of variants at ≥1% MAF and ≥5 supporting reads was 19 in dual 9 dpi, 21 in dual 21 dpi, 19 in single 9 dpi, and 22 in single 21 dpi (Table 1). Group medians for the ≥2% and ≥5% thresholds were 6–7 and 0–1, respectively. Thus the qualitative observation is stable: a modest set of low-frequency alleles is present in all treatments, and the bulk of candidate variants sits between 1% and 5%.

Fitting the additive model $y = \beta_0 + \beta_1 \cdot \text{dual} + \beta_2 \cdot 21\ \text{dpi}$ to the 12 library-level values gave a dual-minus-single effect of −0.33 variants (95% CI −1.83 to 1.34; permutation p = 0.79; Figure 2B). The treatment-by-dpi interaction was −0.33 (95% CI −3.33 to 3.00; p = 0.93). These estimates are imprecise because each cell contains three replicates, but they do not support a systematic increase in WSMV minor-variant burden under coinfection.

Nucleotide diversity and substitution spectra

Mean per-site nucleotide diversity (π) over covered WSMV sites was 1.01–1.17×10^{-3} across groups, consistent with the low diversity reported for WSMV [1, 2, 4]. The dual-minus-single effect on π was −1.07×10^{-4} (95% CI −2.07×10^{-4} to −9.6×10^{-6}; permutation p = 0.092). This is the smallest p value among the seven sample-level endpoints, but after Benjamini–Hochberg correction across those seven tests the FDR is approximately 0.65. The bootstrap confidence interval excludes zero while the permutation p value does not cross 0.05; because these procedures make different assumptions and the effect is small, we do not interpret this as evidence of a coinfection effect. Summed π across detected variant sites, which weights the low-frequency tail, showed an effect of −0.027 (95% CI −0.101 to 0.056; p = 0.61; Figure 2C).

Substitution spectra were dominated by transitions in every library (Ts fraction 0.50–0.67; group means 0.53 single versus 0.54 dual). After normalizing each group to 100% of substitutions, the dual-minus-single difference in Ts fraction was +0.011 (permutation p = 0.80; Figure 3A). In separate tests of the 12 substitution classes (400 dpi-stratified permutations, Benjamini–Hochberg across classes), the smallest FDR was 1.0. The aggregate spectrum is therefore similar at the resolution tested, although this does not exclude changes in mutational processes that the spectrum cannot resolve.

Site-level architecture of recurrent variants

To ask whether coinfection changes the frequency of individual variants rather than their count, we restricted analysis to sites with ≥400× depth in at least five libraries of each treatment arm and fitted the same additive model to site-level MAF. Seventy-three candidate sites met this criterion and were tested with 400 dpi-stratified permutations and Benjamini–Hochberg correction. The median effect was −0.00058, 31 of 73 effects were positive, and no site reached FDR < 0.10. The site-level distribution is therefore centered near zero, with no evidence for a subset of strongly responsive sites.

Eleven positions—430, 1624, 2672, 5722, 5941, 6007, 6577, 6671, 8191, 9154, and 9207—were shared at both time points, and 15 sites were shared within each time point (Figure 2A). Four dual-specific and two single-specific sites were observed at 21 dpi, and one of each at 9 dpi, but these counts are descriptive and no site passed site-level FDR < 0.10. If these positions represent true variants rather than recurrent mapping or reference artifacts, one explanation is a variant pool largely shared before or early in infection; inoculum sequencing would be needed to test this hypothesis.

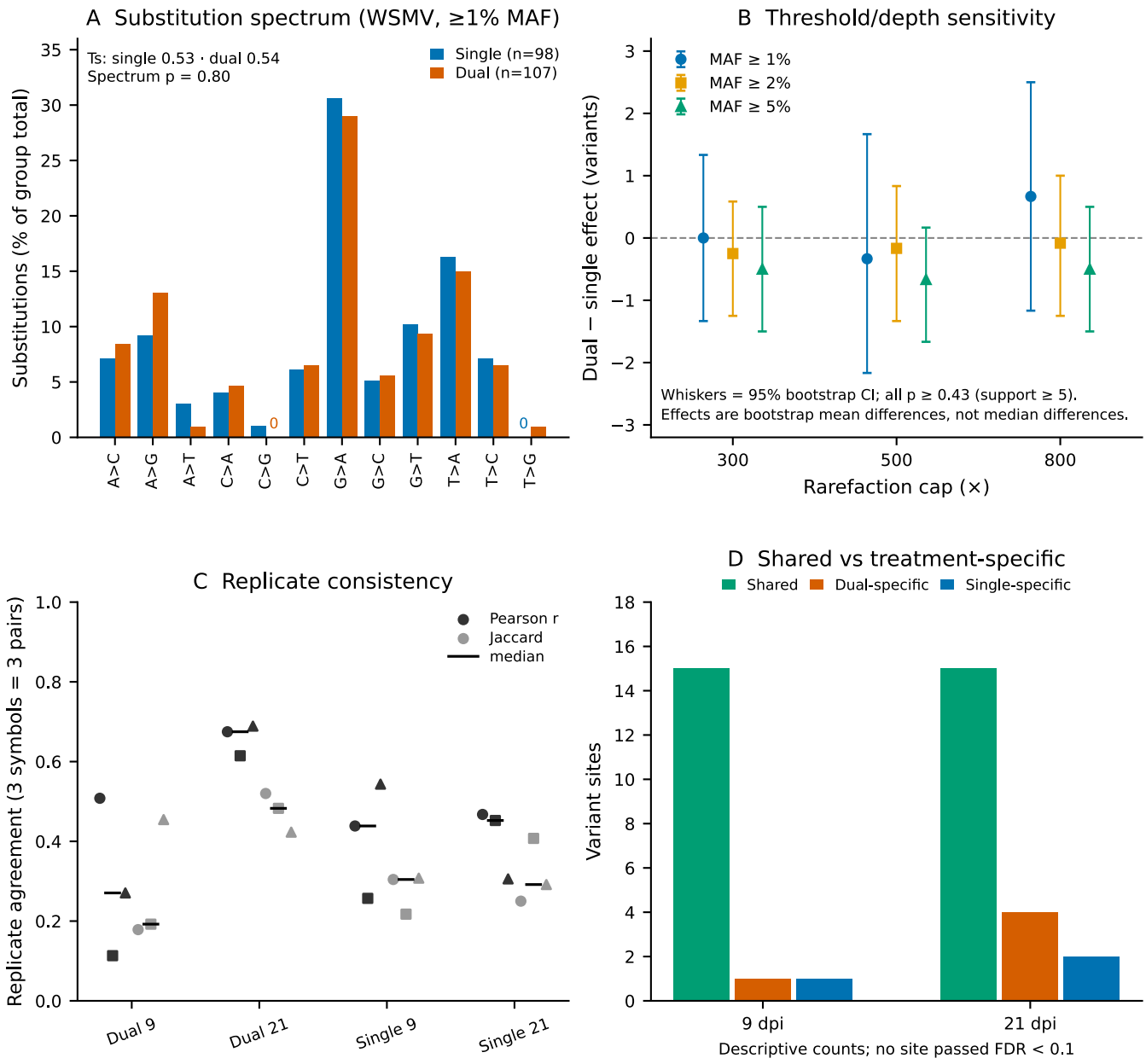


Figure 3. Spectrum, robustness, and replicate concordance. (A) Substitution spectrum for WSMV variants (≥1% MAF, ≥5 reads), normalized to 100% of substitutions within each group; Ts, mean per-library transition fraction; zero-height bars are labeled 0. (B) Dual-minus-single effect on variant counts across rarefaction caps and MAF thresholds; points are bootstrap mean differences and whiskers are 95% bootstrap CIs; the dashed line is zero; all support-≥5 p values are ≥0.43. (C) Pairwise Pearson correlation (dark circles) and Jaccard index (light circles) among the three replicates of each group; marker shape (circle, square, triangle) distinguishes the three replicate pairs; horizontal bars are medians. (D) Shared and treatment-specific WSMV variant sites at 9 and 21 dpi; counts are descriptive because no site passed FDR < 0.10.

Replicate concordance and shared versus specific variants

Within-group agreement was moderate and time-dependent. Pairwise Pearson correlations of site-level MAF ranged from 0.10 to 0.69, and Jaccard indices of the detected variant sets ranged from 0.18 to 0.52 (Figure 3C). The highest agreement was between dual 21 dpi replicates (r = 0.60–0.69), whereas dual 9 dpi replicates were more variable (r = 0.10–0.51). Single-infection groups were intermediate. This heterogeneity is consistent with stochastic variant amplification at low frequency and reinforces the need for matched-depth comparisons and conservative inference.

At 9 dpi, 15 sites were shared between treatments, with one dual-specific and one single-specific site; at 21 dpi, 15 sites were shared, with four dual-specific and two single-specific sites (Figure 3D). For TriMV, three sites were shared between 9 and 21 dpi, with two 9-dpi-specific and one 21-dpi-specific site; these counts come from the TriMV pileups, not from the WSMV

tables. Because the inoculum was not sequenced, we cannot distinguish variants inherited from the inoculum from those generated de novo after infection; recurrent artifacts could also contribute.

Sensitivity to analytical choices

We repeated the variant-count comparison across rarefaction caps of 300×, 500×, and 800×; MAF thresholds of 1%, 2%, and 5%; and read-support cutoffs of 1 and 5. Across the 18 combinations, permutation p values ranged from 0.43 to 1.00; for the primary support-5 grid, all p values were ≥0.43. Effect estimates changed sign across combinations and their 95% bootstrap intervals consistently included zero (Figure 3B). The absence of a robust positive effect is therefore not an artifact of a single depth or threshold choice.

Discussion

We asked whether TriMV coinfection detectably changes the within-host diversity of WSMV. Using 12 libraries from a common host genotype and controlled inoculation, thinned to an expected 500× depth cap, we found that WSMV minor variants are abundant in all treatments but that the estimated coinfection effect is small, imprecise, and not statistically significant for any sample-level endpoint after correction. Site-level tests similarly produced a distribution centered near zero, with no site passing FDR < 0.10. The result was stable across rarefaction depths, MAF thresholds, and read-support cutoffs. These data do not establish that coinfection has no effect; they fail to detect one of the size this design can resolve.

These findings do not contradict reports that coinfection can increase WSMV accumulation [3, 5]. Read fractions and depth are compositional measures, and a higher number of viral templates need not increase the proportional representation of low-frequency alleles. Several non-exclusive hypotheses could account for the absence of a detected shift: a larger template pool could increase the absolute number of mutant genomes while leaving their frequency distribution unchanged; fitness costs or tight bottlenecks could constrain variant frequencies; or the compared populations could share most variants before treatment divergence. Distinguishing these possibilities would require absolute quantification of viral genomes and sequencing of the inoculum, neither of which is available here.

The recurrence of the same variant positions across treatments is notable. Shared sites at 430, 1624, 2672, 5722, 5941, 6007, 6577, 6671, 8191, 9154, and 9207 appeared at both time points, and 15 sites were shared within each time point. If these are true variants, one explanation is a common founder pool rather than independent de novo generation under each treatment; recurrent mapping or reference artifacts remain an alternative that this dataset cannot exclude. It also means that the relevant comparison is the frequency of shared variants, not the appearance of new positions, a point made previously for WSMV population studies [1, 2]. Our site-level analysis was designed accordingly and found no frequency shift.

For studies that use WSMV diversity as a marker of mixed infection, these results suggest caution: in this experiment, the presence of TriMV did not systematically inflate WSMV minor-variant burden. Diversity estimates from field samples may therefore reflect inoculum composition, vector transmission bottlenecks, and time since infection more than the presence of TriMV, although this dataset cannot rank those contributions. The confidence intervals are wide enough to accommodate modest effects, and the analysis has limited power with three replicates per cell. A larger design, absolute viral quantification, and inoculum sequencing would be needed to exclude small but biologically meaningful changes.

Limitations of the study

- Only 12 libraries (three replicates per cell) were analyzed, so power to detect small effects is limited; confidence intervals are correspondingly wide, and no equivalence margin was prespecified.
- This 12-library subset does not include the nine TriMV-only runs available in the public project (PRJNA1165236), so all comparisons are WSMV-centric; we cannot test whether WSMV affects TriMV diversity. Including that arm was a scope choice, not a limitation of the source data.
- The inoculum was not sequenced, so inherited and de novo variants cannot be distinguished.

- The consensus reference was built from the same data; although only one fixed difference was found and incorporated, reference choice can affect low-frequency calls.
- Twenty-seven WSMV site–sample instances (40 including TriMV) met the strand-bias flag. A sensitivity analysis excluding those positions gave a variant-count effect of −1.00 (95% CI −2.83 to 1.00; p = 0.54) and a π effect of -1.10×10^{-4} (95% CI -2.20×10^{-4} to -1.52×10^{-5}; p = 0.092), so the conclusions were unchanged.
- Read fractions and depth are compositional and should not be read as absolute viral loads. The non-reference base background rate includes true low-frequency variants, RT/PCR errors, and mapping effects; it is not an independent measurement of sequencing error. No RT-qPCR or plaque validation of minor variants was performed.
- Results pertain to wheat cv. Tomahawk under controlled conditions and may not transfer to other cultivars or field populations.

STAR★Methods

Key resources table

Resource	Source / identifier	Use
Wheat RNA-seq libraries (12 of 36 runs)	NCBI BioProject PRJNA1165236; Nunna et al. [7]	Viral read mapping and variant calling
WSMV Sidney 81 reference	GenBank AF057533	Reference for consensus construction
TriMV Nebraska reference	GenBank FJ669487	Reference for consensus construction
Consensus reference	This study; TriMV 8233 T→A incorporated	Mapping and pileup
fastp 1.4.0	Open-source	Adapter and quality trimming
BWA 0.7.19-r1273	Open-source	Read mapping
SAMtools 1.25	Open-source	Sorting, fixmate, duplicate marking, indexing
bcftools 1.25	Open-source	mpileup and multiallelic calling
pysam 0.24.1	Open-source	Pileup and depth summaries
NumPy 2.5.4, pandas 3.1.0, matplotlib 3.11.2	Open-source	Analysis and figures
OpenBioWorkstation + Snakemake 9.27.1	This repository	Workflow orchestration, caching, and UI demonstration

Resource availability

Lead contact. Not applicable; this is an automated analysis with no individual corresponding author.

Materials availability. No new materials were generated.

Data and code availability. Raw reads are available under NCBI BioProject PRJNA1165236. Processed tables are in the study `analysis/` directory; scripts for download, consensus construction, workflow generation, variant analysis, figures, and PDF assembly are in `scripts/`. No new deposited accession was created.

Software review status. The external bcftools auto-wrap definition carries `reviewed: true`. That flag was set during Phase B by the automated agent acting as the user's proxy, as instructed in the Phase B task file. It does not represent review by an independent human, and it conflicts with the project brief's requirement for human review before installation. Phase D did not change the flag or claim human approval; a genuine human review record would still be required if that acceptance criterion is enforced.

Experimental model and subject details

The original experiment used wheat cv. Tomahawk, inoculated with WSMV Sidney 81 alone or together with TriMV Nebraska, and sampled upper uninoculated leaves at 9, 12, and 21 dpi with three biological replicates per treatment [7]. We used the 9 and 21 dpi samples of the WSMV and dual-infection arms (12 libraries). Library preparation and sequencing were performed by the original authors; we performed only computational subsampling and reanalysis.

Method details

Read subsampling and quality control

For each library, 500,000 read pairs were streamed from the SRA using `fastq-dump -X 500000 --split-files --gzip` via the NCBI AWS mirror; read counts were verified with `seqkit stats`, with up to five retries on failure. The 24 formal-download FASTQ files totaled 890,305,038 bytes (compressed size on disk). The pilot FASTQ used for consensus construction added 118,491,875 bytes, and all study FASTQ files together totaled 1,008,796,913 bytes. These are file sizes, not measured network traffic; the raw directory occupied 1,009,579,217 bytes including metadata. Reads were trimmed with fastp 1.4.0 using default adapter detection and quality settings; the resulting Q30 rate was ≥95.6% in all libraries and the duplicate rate was 10.0–14.6% (Figure 1C).

Consensus reference

Pilot pileups from four WSMV and two TriMV libraries were used to build a consensus for each virus (depth ≥100, total depth ≥200, major-allele frequency ≥0.8). The WSMV consensus matched AF057533; the TriMV consensus differed from FJ669487 at position 8233 (T→A; depth 1,215; frequency 1.0000). This difference was incorporated into the combined reference used for mapping, so that the variant was not called as a fixed difference in every library.

Mapping, deduplication, and pileup

Reads were mapped with BWA MEM 0.7.19-r1273 using default parameters. Unmapped reads were removed with `samtools view -F 4`; reads were name-sorted, processed with `fixmate -m`, coordinate-sorted, and duplicates were marked with `samtools markdup -s`. Pileup used `stepper="all"` with flag filter 0xF04 (unmapped, secondary, QC-fail, duplicate, supplementary), minimum base quality 30, explicit per-read minimum mapping quality 20 (applied because `stepper="all"` does not enforce the pysam mapping-quality argument), and htslib overlap clipping (`ignore_overlaps=True`); BAQ was not applied. Per-site counts of A, C, G, T, and N, plus reference- and alternate-strand counts, were converted to depth, major non-reference allele, MAF, and per-site π. The non-reference base background rate was estimated as the total non-reference base count divided by total base count at covered positions; it is not a pure sequencing-error estimate.

Expected-depth rarefaction

To reduce the dependence of variant detection on depth, each base count was thinned independently with a binomial draw using $p = \min(1, \text{cap}/\text{depth})$, so that sites at or above the cap had an expected depth of 500× while sites below the cap were left unchanged (100 replicates). This is expected-depth thinning, not fixed-size sampling; the median rarefied site depth was 499–500×, and 1.54% of raw candidate site observations remained below the cap. For each replicate, variant counts and diversity metrics were recomputed, and the median across replicates was used as the library-level value. Sensitivity analyses used caps of 300×, 500×, and 800× with 50 replicates each.

Diversity and variant endpoints

π was computed per site as $1 - \Sigma p_i^2$, corrected by $n/(n-1)$, and averaged over covered sites (π_{all}) or summed over detected variant sites ($\Sigma\pi_{detected}$). Minor variants were defined as non-reference alleles with MAF ≥1%, ≥2%, or ≥5% and at least 1 or 5 supporting reads, excluding sites above the fixed threshold. A site was recurrent if it reached ≥1% MAF in at least 2 of 12 libraries. Replicate agreement was measured by Pearson correlation of site-level MAF and by Jaccard index of detected variant sets.

Strand bias and overlap

For sites with depth ≥100, MAF ≥1%, and at least 5 major-alt reads, a two-sided Fisher exact test compared the reference and alternate allele counts by read strand (2×2 table). Sites with $p < 0.01$ and $|f_{alt} - f_{ref}| > 0.2$ were flagged. Forty site–sample instances were flagged across both viruses, including 27 WSMV instances. A sensitivity analysis re-ran the three primary sample-level endpoints after excluding flagged WSMV positions from each affected library. Paired-end overlap was clipped by htslib during pileup; the coordinate-based overlap coefficient (median 0.88–0.89 across libraries) was recorded as a diagnostic only and was not applied a second time to depth.

Quantification and statistical analysis

Sample-level endpoints were analyzed with the additive linear model $y = \beta_0 + \beta_1 \cdot \text{dual} + \beta_2 \cdot 21\ \text{dpi} + \varepsilon$; the effect of interest is β_1. Two-sided p values were obtained by permuting treatment labels within each dpi stratum (3 of 6 libraries labelled dual; 400 combinations) and comparing the absolute observed effect to the null distribution. Confidence intervals were obtained from 2,000 stratified bootstrap resamples (resampling within each of the four treatment-by-dpi cells) and are reported as percentile intervals. The seven sample-level endpoints were corrected by the Benjamini–Hochberg procedure.

Site-level effects were the additive dual-versus-single coefficient from $y = \beta_0 + \beta_1 \cdot \text{dual} + \beta_2 \cdot 21\ \text{dpi}$ fitted to site-level MAF across the 12 libraries. p values came from 400 dpi-stratified treatment-label permutations (20 × 20), with the same +1 correction as the sample-level tests; the 73 candidate sites were corrected by Benjamini–Hochberg. The aggregate Ts-fraction test and the 12 individual substitution-class tests used the same stratified permutation scheme and BH correction. Because p values are computed from 400 enumerated permutations, their resolution is $1/401 \approx 0.0025$ rather than continuous; the +1 correction avoids reporting an exact zero. Robustness analyses repeated the variant-count comparison across 3 rarefaction caps × 3 MAF thresholds × 2 support cutoffs. The interaction test uses the same label permutations and assumes exchangeability under the additive null; when main effects are non-zero, its permutation distribution is approximate, so the interaction p value is descriptive. No test was one-sided; $\alpha = 0.05$ before correction and FDR < 0.10 for site-level screening. Analyses used NumPy 2.5.4 and pandas 3.1.0; figures used matplotlib 3.11.2.

Figure legends

Figure 1. Viral read composition and sequencing depth. (A) WSMV and TriMV read pairs per library; percentages are of 500,000 total pairs. (B) ECDF of effective WSMV depth; inset values are median breadth at 100×, 400×, and 1,000×. (C) Duplicate rate versus insert-size peak. Read fractions are compositional, not absolute viral loads.

Figure 2. Recurrent WSMV minor variants and sample-level effects. (A) Heatmap of 31 recurrent sites; light gray, MAF <1%; values are medians over 100 thinning replicates at a 500× cap. (B) Variants at ≥1% MAF with ≥5 reads per library. (C) Summed π across detected variant sites. Effect, 95% bootstrap CI, permutation p, and FDR are from the seven-endpoint family. Replicate numbers are not paired across treatments, so no connecting lines are shown.

Figure 3. Spectrum, robustness, and replicate concordance. (A) Substitution spectrum for WSMV variants (≥1% MAF, ≥5 reads), normalized to 100% of substitutions within each group; Ts, mean per-library transition fraction; zero-height bars are labeled 0. (B) Dual-minus-single effect on variant counts across rarefaction caps and MAF thresholds; points are bootstrap mean differences and whiskers are 95% bootstrap CIs; the dashed line is zero; all support-≥5 p values are ≥0.43. (C) Pairwise Pearson correlation (dark circles) and Jaccard index (light circles) among the three replicates of each group; marker shape (circle, square, triangle) distinguishes the three replicate pairs; horizontal bars are medians. (D) Shared and treatment-specific WSMV variant sites at 9 and 21 dpi; counts are descriptive because no site passed FDR < 0.10.

References

1. French R, Stenger DC. Population structure within lineages of Wheat streak mosaic virus derived from a common founding event exhibits stochastic variation inconsistent with the deterministic quasi-species model. *Virology*. 2005;343(2):179–189. doi:10.1016/j.virol.2005.08.037.
2. French R, Stenger DC. Evolution of Wheat streak mosaic virus: dynamics of population growth within plants may explain limited variation. *Annu Rev Phytopathol*. 2003;41:199–214. doi:10.1146/annurev.phyto.41.052002.095559.
3. Tatineni S, Graybosch RA, Hein GL, Wegulo SN, French R. Wheat cultivar-specific disease synergism and alteration of virus accumulation during co-infection with Wheat streak mosaic virus and Triticum mosaic virus. *Phytopathology*. 2010;100(3):230–238. doi:10.1094/PHYTO-100-3-0230.

4. Bartels M, French R, Graybosch RA, Tatineni S. Triticum mosaic virus exhibits limited population variation yet shows evidence of parallel evolution after replicated serial passage in wheat. *Virology*. 2016;492:92–100. doi:10.1016/j.virol.2016.02.010.

5. Tatineni S, Alexander J, Qu F. Differential synergistic interactions among four different wheat-infecting viruses. *Front Microbiol*. 2021;12:800318. doi:10.3389/fmicb.2021.800318.

6. Redila CD, Phipps S, Nouri S. Full genome evolutionary studies of Wheat streak mosaic-associated viruses using high-throughput sequencing. *Front Microbiol*. 2021;12:699078. doi:10.3389/fmicb.2021.699078.

7. Nunna H, Palmer NA, Sarath G, Wegulo SN, Tatineni S. Synergistic interaction between Wheat streak mosaic virus and Triticum mosaic virus modulates wheat transcriptome to favor disease severity. *Front Plant Sci*. 2024;15:1504482. doi:10.3389/fpls.2024.1504482.

8. Ranabhat NB, Fellers JP, Bruce MA, Rupp JLS. High-throughput Oxford Nanopore sequencing unveils complex viral population in Kansas wheat: implications for sustainable virus management. *Viruses*. 2025;17(1):126. doi:10.3390/v17010126.

9. Li D, Song X, Yang F. Global distribution, evolutionary dynamics, and origins of Wheat streak mosaic virus. *Front Plant Sci*. 2025;16:1611008. doi:10.3389/fpls.2025.1611008.

10. Gautam S, Gadhave KR. Field evidence for asymmetric regulation of Wheat streak mosaic virus and Triticum mosaic virus across the wheat–wheat curl mite interface. *Insects*. 2026;17(5):459. doi:10.3390/insects17050459.

中文摘要

标题：小麦条纹花叶病毒与小麦花叶病毒共感染对 WSMV 群体内多样性的影响：基于期望深度稀疏化的证据

作者：OpenBioWorkstation 课题组（自动化分析）

摘要：小麦条纹花叶病毒（WSMV）与小麦花叶病毒（TriMV）常在同一麦株上混合感染，但共感染是否改变 WSMV 的群体内多样性仍不清楚。本研究从公共数据 PRJNA1165236 中选取 12 个 RNA-seq 文库，构成 2×2 设计（单感染/双感染 × 9/21 天 × 3 个重复），每个样本取 50 万对 reads，比对到数据共识参考序列，去除重复后按期望深度二项稀疏化到 500× 上限，再计算核苷酸多样性 π、低频变异数与替换频谱。结果显示，各组均存在大量低频变异（≥1% 且 ≥5 条支持 reads 的组中位数为 19–22 个/样本）；双感染减单感染的效应为 −0.33 个变异（95% CI −1.83 至 1.34，置换检验 $p = 0.79$），交互项 −0.33（−3.33 至 3.00，$p = 0.93$）。π、替换频谱与位点水平等位频率在多重校正后均无显著差异；73 个受检候选位点中 0 个达到 FDR < 0.10（其中 31 个为复发位点）。上述结果在 300×/500×/800× 稀疏化上限、1%/2%/5% 阈值与 1/5 条支持 reads 的 18 种组合下保持一致（$p \geq 0.43$）。因此，在这批样本中未检测到 TriMV 共感染显著改变 WSMV 群体内多样性的证据；每组 3 个重复的功效有限，不能据此证明无效应。

关键词：小麦条纹花叶病毒；小麦花叶病毒；混合感染；群体内多样性；低频变异；RNA 测序；稀疏化

说明：本页为中文摘要，英文正文与图表见前文。所有数字来自 analysis/ 下的实测输出；reads 比例与深度为组成性指标，不表示绝对病毒载量。